Practice Set 5

종합 응용 · Final Challenge

Name: _____ Student ID: _____

Total: 100 points | Suggested time: 80-95 minutes | Part A: 20 · Part B: 30 · Part C: 50

Instructions: Answer the multiple-choice questions, justify true/false responses, and show all key calculations in written answers. Use $\Phi(1)=0.8413$, $\Phi(1.5)=0.9332$, $\Phi(2)=0.9772$, $e^{-1}\approx0.3679$, $e^{-2}\approx0.1353$, $e^{-3}\approx0.0498$ as needed.

PART A · MULTIPLE CHOICE (10 × 2 = 20 pts)

MC1. P(D)=0.30, P(+|D)=0.80, and P(+|D)=0.10. Find P(D|+).

(a) 0.3000

(b) 0.5333

(c) 0.7742

(d) 0.9000

MC2. X ~ Bernoulli(0.20) and Y=5X 2. Find Var(Y).

(a) 0.16

(b) 0.80

(c) 2.00

(d) 4.00

MC3. X ~ Binomial(10, 0.10). What is Var(X)?

(a) 0.09

(b) 0.90

(c) 1.00

(d) 9.00

MC4. Arrivals are Poisson at 2 per hour. Find P(no arrivals in 30 minutes).

(a) e^(4)

(b) e^(2)

(c) e^(1)

(d) 1 e^(1)

MC5. X ~ Binomial(4, 0.5). Find P(X≥3).

(a) 1/16

(b) 1/4

(c) 5/16

(d) 1/2

MC6. X and Y are independent with sd(X)=3 and sd(Y)=4. Find Var(2X Y).

(a) 14

(b) 25

(c) 52

(d) 100

MC7. For an integer-valued X, F(2)=0.65 and F(1)=0.40. Find P(X=2).

(a) 0.15

(b) 0.25

(c) 0.40

(d) 0.65

MC8. Which Bernoulli probability p yields the largest variance?

(a) 0.1

(b) 0.3

(c) 0.5

(d) 0.9

MC9. T ~ Normal(50, 8²). Find P(T>62) using Φ(1.5)=0.9332.

(a) 0.0668

(b) 0.1587

(c) 0.4332

(d) 0.9332

MC10. T ~ Exponential(rate=0.25 per minute). Find P(T>8).

(a) 0.0183

(b) 0.1353

(c) 0.2500

(d) 0.8647

PART B · TRUE / FALSE (10 × 3 = 30 pts)

TF1. Two tests can be conditionally independent given disease status without being independent unconditionally.

○ True ○ False Explain: _____

TF2. If two independent Bernoulli variables have different success probabilities, their sum is always Binomial(2,p) for some p.

○ True ○ False Explain: _____

TF3. A CDF is a nondecreasing function with values between zero and one.

○ True ○ False Explain: _____

TF4. A Bernoulli(0.5) random variable has variance 0.5.

○ True ○ False Explain: _____

TF5. Cov(X,Y)=0 is enough to prove independence in general.

○ True ○ False Explain: _____

TF6. For a continuous Uniform(a,b) variable, P(X=c)=0 for c inside (a,b).

○ True ○ False Explain: _____

TF7. The coefficient of variation is generally problematic when the mean is zero.

○ True ○ False Explain: _____

TF8. If tests are independent conditional on D, $P(+,+|D)=P(+|D)^2$ for identical tests.

○ True ○ False Explain: _____

TF9. For X~Poisson(λ), the standard deviation equals λ.

○ True ○ False Explain: _____

TF10. For X~Binomial(5,0.2), P(X=5)=0.2 .

○ True ○ False Explain: _____

PART C · SHORT ANSWERS (50 pts)

C1. A farm independently checks 12 irrigation controllers. Each has probability 0.10 of failure. Let X be the number of failed controllers.

(a) [5 pt] Find P(X=2).

(b) [5 pt] Find P(X≥1).

(c) [5 pt] Find E(X), Var(X), and sd(X).

C2. Crop disease prevalence is P(D)=0.04. Each diagnostic test has sensitivity P(+|D)=0.92 and false-positive rate P(+|D)=0.07. Two tests are conditionally independent given disease status.

(a) [5 pt] Find P(+) for the first test.

(b) [5 pt] Find P(D|+) after one positive test.

(c) [5 pt] Find P(D|+,+) after two positive tests.

C3. A food-producer's net operating return X (thousand KRW) has PMF: x= 30,10,50,90 with probabilities 0.1,0.4,0.3,0.2. Let Y=1.5X+10. Independent input cost C has E(C)=8 and sd(C)=6; total net profit Z=1.5X C+10.

(a) [5 pt] Compute E(X), E(X²), Var(X), and sd(X).

(b) [5 pt] Compute E(Y), Var(Y), and sd(Y).

_____

_____

(c) [5 pt] Compute E(Z), Var(Z), and sd(Z), showing why independence matters.

_____

_____

(d) [5 pt] Find $P(X \geq 50)$ and interpret it.

_____

_____

ANSWER KEY & DETAILED SOLUTIONS

정답과 자신의 풀이를 비교하고, 문제에서 사용한 확률분포의 가정도 점검하세요.

PART A · 정답 및 해설

MC1 C MC2 D MC3 B MC4 C MC5 C MC6 C MC7 B MC8 C MC9 A MC10 B

MC1 · C

P(D|+)=.30×.80/[.30×.80+.70×.10]=.24/.31≈.7742.

MC2 · D

Var(Y)=5²×p(1 p)=25×.2×.8=4.

MC3 · B

Var(X)=10×.1×.9=.9.

MC4 · C

30분 평균 λ=1이므로 P(0)=e^(1).

MC5 · C

P(3)+P(4)=[C(4,3)+C(4,4)]/2⁴=5/16.

MC6 · C

Var(2X Y)=4·9+16=52. 독립이면 공분산은 0이다.

MC7 · B

P(X=2)=F(2) F(1)=.65 .40=.25.

MC8 · C

p(1 p)의 최대값은 p=.5일 때이다.

MC9 · A

Z=(62 50)/8=1.5; P(Z>1.5)=1 .9332=.0668.

MC10 · B

생존확률=e^(.25×8)=e^(2)≈.1353.

PART B · True / False 해설

TF1 T TF2 F TF3 T TF4 F TF5 F TF6 T TF7 T TF8 T TF9 F TF10 T

TF1 · TRUE

공통 원인인 질병 상태를 혼합하면 검사결과 간 의존이 생길 수 있다.

TF2 · FALSE

일반적인 이항분포는 두 시행에서 같은 p를 요구한다. p가 다르면 포아송이항분포가 된다.

TF3 · TRUE

CDF는 누적확률이므로 x가 증가해도 감소할 수 없다.

TF4 · FALSE

분산은 .5×.5=.25이다.

TF5 · FALSE

상관이 0이어도 비선형 의존성이 존재할 수 있다.

TF6 · TRUE

연속확률변수의 한 점에 대한 확률은 0이다.

TF7 · TRUE

CV=s/mean이므로 평균 0이면 정의되지 않는다.

TF8 · TRUE

동일한 조건부 성공확률과 조건부독립성이 함께 성립해야 한다.

TF9 · FALSE

표준편차는 $\sqrt{\lambda}$이고, λ와 같은 것은 분산이다.

TF10 · TRUE

5번 모두 성공할 확률은 C(5,5)×.2 ×.8 =.2 이다.

PART C · 단계별 서술형 풀이

C1 · A farm independently checks 12 irrigation controllers. Each has probability 0.10 of failure. Let X be the number of failed controllers.

(a) Find P(X=2).

X~Binomial(12,.1). P(X=2)=C(12,2)×.1²×.9¹⁰=66×.01×.34867844≈0.23013.

(b) Find P(X≥1).

P(X≥1)=1 .9¹²≈1 .28242954=.71757.

(c) Find E(X), Var(X), and sd(X).

E(X)=12(.1)=1.2, Var(X)=12(.1)(.9)=1.08, sd(X)=√1.08≈1.0392.

C2 · Crop disease prevalence is P(D)=0.04. Each diagnostic test has sensitivity P(+|D)=0.92 and false-positive rate P(+|D)=0.07. Two tests are conditionally independent given disease status.

(a) Find P(+) for the first test.

전확률: P(+)=.04(.92)+.96(.07)=.0368+.0672=.1040.

(b) Find P(D|+) after one positive test.

베이즈: P(D|+)=.0368/.104≈.35385.

(c) Find P(D|+,+) after two positive tests.

조건부 독립성으로 분자는 .04×.92²=.033856, 비감염 경로는 .96×.07²=.004704.
후행확률=.033856/(.033856+.004704)=.033856/.03856≈.87801.

C3 · A food-producer's net operating return X (thousand KRW) has PMF: x= 30,10,50,90 with probabilities 0.1,0.4,0.3,0.2. Let Y=1.5X+10. Independent input cost C has E(C)=8 and sd(C)=6; total net profit Z=1.5X C+10.

(a) Compute E(X), E(X²), Var(X), and sd(X).

E(X)= 30(.1)+10(.4)+50(.3)+90(.2)= 3+4+15+18=34. E(X²)=900(.1)+100(.4)+2500(.3)+8100(.2)=2500.
Var(X)=2500 34²=1344, sd≈36.6606.

(b) Compute E(Y), Var(Y), and sd(Y).

E(Y)=1.5×34+10=61. Var(Y)=1.5²×1344=3024, sd(Y)=√3024≈54.9909.

(c) Compute E(Z), Var(Z), and sd(Z), showing why independence matters.

E(Z)=1.5×34 8+10=53. Var(Z)=1.5²×1344+Var(C) 3Cov(X,C)=3024+36=3060 (독립 → Cov=0).
sd(Z)=√3060≈55.3173.

(d) Find P(X≥50) and interpret it.

P(X≥50)=P(X=50)+P(X=90)=.3+.2=.5. 운영수익이 50천원 이상일 확률이 50%다.

계산 점검: ① 확률의 합=1 ② 조건부 사건 분모 확인 ③ 독립성 확인 ④ E(X²)와 [E(X)]² 구별 ⑤ 선형변환 시 분산 계수 제곱 ⑥ 관측 구간과 포아송 λ 일치